MASTER CLINICAL SUPPLY QUALITY AGREEMENT (Q-AGR)

Document Ref: BioMarin High-Titer AAV Quality Agreement (Q-AGR-BMR-19) | **Standard:** FDA 21 CFR Part 11 & EU GMP Annex 15

1. Commercial Parties & Asset Specifications

Commercial Sponsor	BioMarin Pharmaceutical
Sponsor Identifier	CUST-BMRN-06
Therapeutic Asset	**Roctavian (valoctocogene roxaparvovec) AAV5 Gene Therapy**
Active Batch / Lot #	LOT-BMR-AAV-3301
Commercial Valuation	**$620,000 USD**
Consignee Medical Facility	Johns Hopkins Hospital Cellular Suite

2. Storage Specifications & Excursion Limits

The carrier is legally and operationally bound to preserve custody within qualified physical boundaries under Tier **Transit Vibration & Mechanical G-Shock SLA (SLA-MECHANICAL-GFORCE)**. Continuous edge telemetry samples temperature at ≤ 1.0 second frequency with cryptographically verifiable records.

Qualified Temperature Band	**+2.0°C to +8.0°C**
Critical Regulatory Ceiling	8.0°C
Critical Regulatory Floor	2.0°C
Allowable Cumulative Excursion	120 minutes maximum
Sampling & Seal Frequency	Real-Time Telematics (1.0 sample/s)

3. Financial Liability & Liquidated Damages Schedule

Failure to remediate an active physical breach beyond 8.0°C within 120 minutes constitutes complete batch degradation, triggering full commercial reimbursement of **$620,000 USD**. For delays exceeding contract cutoff without thermal breach, liquidated damages accrue at **$60,000 clinical rescheduling fee + G-force audit charge**.

4. Autonomous Triage & Emergency Diversion Protocols

Under FDA 21 CFR Part 11 electronic control provisions, the commercial sponsor confers autonomous executive authority to **CryoGuard AI™**. In the event of primary equipment failure, the system is authorized to:

- Perform multi-criteria facility scoring (distance, bay capacity, certified capabilities).
- Execute closed-loop highway diversion to the nearest compatible secondary biorepository.
- Automatically generate cryptographically signed CAPA non-conformance reports.

5. Regulatory Certification & Electronic Signatures

For Commercial Sponsor: BioMarin Pharmaceutical *Digitally Signed: e-Sig SHA-256* Date: 2026-10-14	**For Logistics Carrier & AI Operator:** CryoGuard Autonomous Logistics Inc. *Automated Execution Authority Active* Date: 2026-10-14